string theory, six dimensions were curled up, while seven were curled up for the eleven-dimensional theory.

When string theory took physicists by storm in 1984, the perspective on point-particle supergravity theories changed dramatically. As emphasized repeatedly, if we examine a string with the precision available currently and for the foreseeable future, it looks like a point particle. We can make this informal remark precise: When studying low-energy processes in string theory—those processes that do not have enough energy to probe the ultramicroscopic, extended nature of the string—we can approximate a string by a structureless point particle, using the framework of point-particle quantum field theory. We cannot use this approximation when dealing with short-distance or high-energy processes because we know that the extended nature of the string is crucial to its ability to resolve the conflicts between general relativity and quantum mechanics that a point-particle theory cannot. But at low enough energies—large enough distances—these problems are not encountered, and such an approximation is often made for the sake of calculational convenience.

The quantum field theory that most closely approximates string theory in this manner is none other than ten-dimensional supergravity. The special properties of ten-dimensional supergravity discovered in the 1970s and 1980s are now understood to be low-energy relics of the underlying power of string theory. Researchers studying ten-dimensional supergravity had uncovered the tip of a very deep iceberg —the rich structure of superstring theory. In fact, it turns out that there are four different ten-dimensional supergravity theories that differ in details regarding the precise way in which supersymmetry is incorporated. Three of these turn out to be the low-energy point-particle approximations to the Type IIA string, the Type IIB string, and the Heterotic-E string. The fourth gives the the low-energy point-particle approximation to both the Type I string and the Heterotic-O string; in retrospect, this was the first indication of the close connection between these two string theories.

This is a very tidy story except that eleven-dimensional supergravity seems to have been left out in the cold. String theory, formulated in ten dimensions, appears to have no room for an eleven-dimensional theory. For a number of years, the general view held by